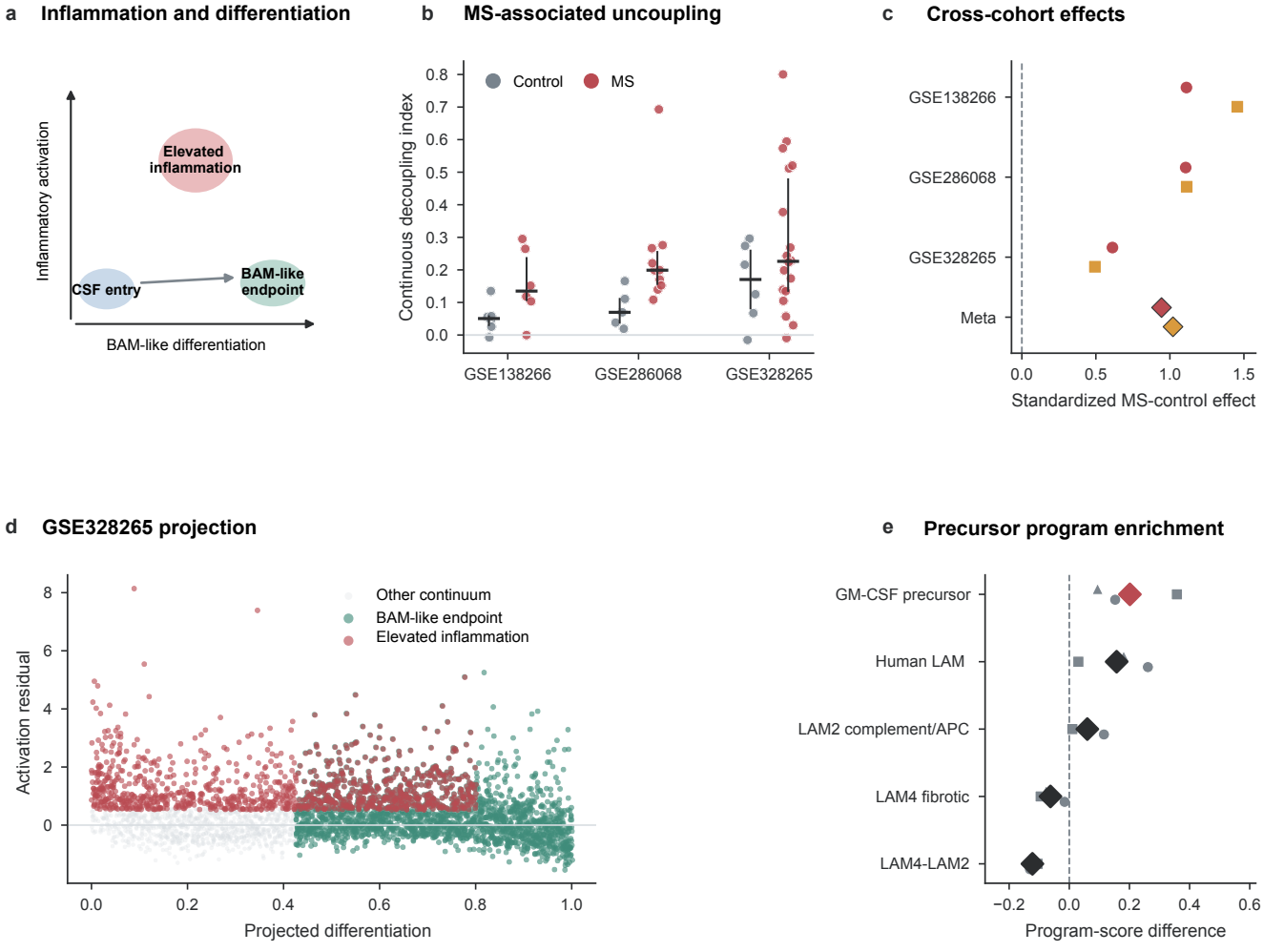

a
Inflammation and differentiation
Inflammatory activation
Elevated inflammation
CSF entry
BAM-like endpoint
BAM-like differentiation
b
MS-associated uncoupling
Control
MS
Continuous decoupling index
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
GSE138266
GSE286068
GSE328265
c
Cross-cohort effects
GSE138266
GSE286068
GSE328265
Meta
0.0
0.5
1.0
1.5
Standardized MS-control effect
d
GSE328265 projection
Other continuum
BAM-like endpoint
Elevated inflammation
Activation residual
8
6
4
2
0
0.0
0.2
0.4
0.6
0.8
1.0
Projected differentiation
e
Precursor program enrichment
GM-CSF precursor
Human LAM
LAM2 complement/APC
LAM4 fibrotic
LAM4-LAM2
−0.2
0.0
0.2
0.4
0.6
Program-score difference